

Renormalization Obstructions in Congruence and Homology Towers of Geodesic Flows

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28 August 2026

Abstract

Finite covers of a hyperbolic surface contain many closed geodesics, but their inverse limit need not possess any periodic point for the coordinatewise geodesic flow. We formalize this owner mismatch and determine what renormalization is required to recover a fixed finite Euler panel. For a descending normal finite-index tower with trivial intersection, every infinite-order owner's quotient orders divide forward and tend to infinity; the inverse-limit flow has no periodic points, and each fixed-owner Euler factor becomes 1 in every fixed coefficient prefix under the physical clock. We specialize the argument to the cusped tower $\Gamma(3n!)$ with an explicit projective-sign proof and compare it with a closed genus-two residual/homology tower, where primitive homology gives factorial lower bounds for minimal lifted periods. We then register a distinct pure homology-cover calibrator $H_N = \ker(\Gamma \rightarrow H_1(\Sigma; \mathbb{Z}/N\mathbb{Z}))$. For a primitive content-one owner, the cover degree, deck order, primitive lift count, and lift period are exactly N^4 , N , N^3 , and $N\ell(g)$. The resulting four-factor theorem shows that physical time alone gives support escape, $1/N$ time rescaling alone gives multiplicity divergence, and $1/N^3$ logarithmic normalization alone still gives support escape. Only their combination recovers $(1 - x_g)^{-1}$ exactly. This positive identity is restricted to a fixed finite panel and is generic for marked genus-two metrics; it is therefore a proves-too-much calibration, not an arithmetic dynamical determinant or a Route-A A2 certificate.

Keywords: inverse-limit flow; residual tower; closed geodesic; quotient-order escape; homology cover; Euler-factor renormalization

繁體中文摘要

有限雙曲曲面覆蓋各自具有許多閉測地線，但這些軌道未必組成逆極限流中的週期點。本文建立一套嚴格的所有者區分，並分析固定有限 Euler 因子面板需要何種重整化。對交集為單位元的下降正規有限指數塔，任一無限階元素在有限商中的階數向前整除且趨於無限；因此座標式測地流沒有週期點，固定所有者的 Euler 因子在物理時間下亦會退出每個固定係數前綴。本文先對尖點塔 $\Gamma(3n!)$ 給出含投影正負號的明確證明，再以封閉 genus-two 殘餘 / 同調塔作控制，得到原始同調所有者之最小提升週期階乘下界。其後另行註冊純同調覆蓋 H_N 。對內容為一的原始所有者，覆蓋次數、deck 階、原始提升分量數與週期分別精確等於 N^4 、 N 、 N^3 與 $N\ell(g)$ 。四象限定理顯示：僅用物理時間會造成支撐逃逸；僅縮放時間會造成重數發散；僅取對數幾何平均仍有支撐逃逸；只有同時施行 $1/N$ 時間縮放與 $1/N^3$ 對數正規化，才在每層精確恢復基底因子。此恆等式只屬於固定有限面板，且對所有標記 genus-two 度量皆成立，故為「證明過多」控制，而非算術動力行列式或 Route-A A2 證書。

關鍵詞：逆極限流；殘餘塔；閉測地線；商階逃逸；同調覆蓋；Euler 因子重整化

1 Introduction

Periodic-orbit constructions require an unambiguous owner. This requirement becomes nontrivial in a tower of finite covers. Each finite cover can have closed geodesics, and a fixed base geodesic can close after a level-dependent number of traversals. A periodic point in the inverse limit, however, requires one coherent point and one common positive time that returns at every coordinate. Confusing these two statements produces an apparent orbit population for a limit flow that may in fact have none.

This paper studies two related questions. The first is an owner-preserving no-go problem: what happens to the periodic orbits and finite Euler factors of a fixed base owner in a normal residual tower with the same physical clock at every level? The second is a controlled rescue problem: after the owner, cover tower, clock, and multiplicity normalization are explicitly changed, which interventions are necessary to recover a fixed finite panel of base Euler factors?

The first answer is rigid. In a descending normal finite-index tower with trivial intersection, the finite quotient order of every infinite-order element tends to infinity. A coherent inverse-limit period would put one fixed nonzero power into every subgroup and is therefore impossible. In physical time, the lifted period of every fixed owner escapes every bounded interval. Algebraically, the corresponding factor $(1 - x_g^{o_n(g)})^{-1}$ converges coefficientwise to 1 on every fixed prefix.

The second answer requires two independent renormalizations. In a pure genus-two homology cover of modulus N , a primitive content-one owner has deck order N , exactly N^3 primitive lifts, and physical period $N\ell(g)$. Time rescaling by $1/N$ repairs the support but leaves a divergent multiplicity N^3 . Dividing the logarithm by N^3 repairs the multiplicity but leaves the physical support at degree N . Only the combined operation returns the base factor exactly.

The article contributes a unified proof chain rather than a broad priority claim. Aperiodic laminated geodesic flows, universal and punctured solenoids, and finite-type hyperbolic solenoidal surfaces already have direct literature. Our contribution is the explicit $\Gamma(3n!)$ specialization with its projective-sign residual proof, a compact-versus-cusped control, a coefficientwise owner firewall, and the four-quadrant theorem for a separately registered finite-panel calibrator. The distinction between those two candidates is maintained throughout.

The Route-A conclusion is negative. The original residual inverse-limit flow has no primitive periodic-orbit population, so it fails the primitive-orbit layer. The fully renormalized homology-panel identity is analytically exact but generic and finite. It has no rational-prime owner map, no full primitive spectrum, and no global dynamical determinant. No prime or Riemann-zero table is used in either construction.

2 Prior work and bounded positioning

Laminated geodesic and horocycle flows are defined leafwise. Martínez, Matsumoto, and Verjovsky give the standard leafwise flow framework, an explicit compact hyperbolic-lamination example without periodic geodesic-flow orbits, and a universal-solenoid example with simply connected leaves. [3, Section 2.2 and Examples 4, 6] These results rule out any claim that general laminated aperiodicity or the simply-connected-leaf mechanism is new here.

Penner and Šarić define the punctured solenoid through finite modular covers and describe its disk leaves. [5, Introduction and Definition 2.1] The directed system in that work differs from the single factorial principal-congruence chain below, but it is direct prior for the noncompact modular-solenoid setting.

Alcalde Cuesta and collaborators define the leafwise geodesic flow and the finite-type hyperbolic solenoidal object class, including inverse limits of regular covers under the McCord terminology. [1, Definitions 4, 5, and 7] Their study concerns horocycle dynamics and does not state the explicit factorial-chain propositions proved below.

For compact weak solenoids, the intersection of a group chain is naturally related to the fundamental group of the corresponding leaf. [2, Definition 5.5] We use this only as structural comparison: the

noncompact congruence case receives a direct group proof. Residual finiteness of the closed surface group follows from the standard theorem that finitely generated linear groups are residually finite. [4, p. 1]

Accordingly, the general mechanism is prior. The exact finite computations and the owner-audit formulation are local contributions, and the negative search for the literal chain $\Gamma(3n!)$ is not treated as proof of novelty.

3 Normal residual towers and the owner firewall

Let Γ be a torsion-free Fuchsian group and

$$\Gamma = \Gamma_1 \supseteq \Gamma_2 \supseteq \cdots$$

a descending sequence of normal finite-index subgroups with $\bigcap_n \Gamma_n = \{e\}$. Put $Y_n = \Gamma_n \backslash \mathbb{H}$ and let

$$M_\infty = \varprojlim_n T^1 Y_n$$

under the tangent maps of the finite coverings. Every coordinate carries the unit-speed geodesic flow, using one hyperbolic-arclength clock.

Definition 3.1 (period owner). A periodic point of M_∞ is a coherent point $x = (x_n)$ and a single $T > 0$ such that $g^T x_n = x_n$ at every level. A sequence of level-dependent closing times T_n is finite-tower data and is not a period of x .

For $g \in \Gamma$, define

$$o_n(g) = \text{ord}(g\Gamma_n \text{ in } \Gamma/\Gamma_n).$$

Theorem 3.2 (residual-tower escape). *The coordinatewise geodesic flow on M_∞ has no periodic point. For every infinite-order $g \in \Gamma$,*

$$o_n(g) \mid o_{n+1}(g), \quad o_n(g) \longrightarrow \infty.$$

Proof. Suppose $x = (\Gamma_n h_n)$ has period $T > 0$. Compatibility with level one gives $h_n = \eta_n h_1$ for some $\eta_n \in \Gamma$. The level-one orbit is a closed geodesic, so there is a primitive hyperbolic $\gamma \in \Gamma$ and an integer $m \geq 1$ such that $T = m\ell(\gamma)$ and

$$h_1 a_T h_1^{-1} = \gamma^m.$$

Return at level n is equivalent to

$$\eta_n \gamma^m \eta_n^{-1} \in \Gamma_n.$$

Normality removes η_n , so $\gamma^m \in \Gamma_n$ for every n . Trivial intersection gives $\gamma^m = e$, contradicting hyperbolicity.

The quotient map $\Gamma/\Gamma_{n+1} \rightarrow \Gamma/\Gamma_n$ implies $o_n(g) \mid o_{n+1}(g)$. If this divisibility sequence were bounded, it would eventually stabilize at some r . Then $g^r \in \Gamma_n$ for every n , so $g^r = e$, contradicting the infinite order of g . $\square$

If g is primitive, its axis stabilizer in a torsion-free Fuchsian group is $\langle g \rangle$. Hence

$$\Gamma_n \cap \langle g \rangle = \langle g^{o_n(g)} \rangle,$$

and the minimal period of each lifted component is $o_n(g)\ell(g)$. Without a primitivity certificate, $o_n(g)\ell(g)$ is only the minimal closing time among whole traversals of the chosen g loop. This terminology prevents finite rows from acquiring stronger orbit credit than their certificates support.

3.1 Quantifiers, coherence, and the role of normality

Theorem 3.2 uses two quantifier changes that are easy to miss. At a fixed level n , a base owner g closes after $o_n(g)$ traversals, so there are many finite-level periodic orbits. A point of the inverse limit is periodic only if there is one time T that closes every coordinate simultaneously. The sequence $o_n(g)\ell(g)$ supplies the former statement and, because it diverges, obstructs the latter. Choosing T after choosing n does not establish a period of the limit flow.

Coherence of the point also matters. A coordinate in the cover may differ from the level-one representative by a conjugator η_n . Return therefore initially places $\eta_n\gamma^m\eta_n^{-1}$, rather than γ^m , in Γ_n . Normality is exactly what removes this level-dependent conjugator. For a nonnormal chain, trivial intersection of the subgroups alone does not justify the same step; one would need control of the conjugates or of an appropriate normal core. The theorem is consequently stated for normal towers and is not silently extended to every inverse system of finite covers.

Residuality has a separate function. It turns membership of one fixed nontrivial power in every level into a contradiction, and it forces the divisibility sequence of quotient orders to be unbounded. Neither finite index nor nestedness alone has this consequence. For example, a chain with nontrivial intersection can retain a closed owner represented by an element of that intersection. Thus the no-period conclusion belongs to the package “normal, descending, finite index, trivial intersection, common clock.” The later homology calibrator deliberately leaves this package and must be treated as a different candidate.

4 The factorial principal-congruence specialization

For $n \geq 1$, let

$$\Gamma_n = \Gamma(3n!) < \text{PSL}_2(\mathbb{Z}).$$

The chain is nested, normal, and finite index. Its residual intersection requires care because principal congruence in the projective group allows a sign.

Lemma 4.1 (projective-sign intersection).

$$\bigcap_{n \geq 1} \Gamma(3n!) = \{1\} \quad \text{in } \text{PSL}_2(\mathbb{Z}).$$

Proof. Let $[A]$ lie in every subgroup and choose $A \in \text{SL}_2(\mathbb{Z})$. For each n there is $\epsilon_n \in \{\pm 1\}$ with

$$A \equiv \epsilon_n I \pmod{3n!}.$$

Reducing the level- $n + 1$ congruence modulo $3n!$ shows $\epsilon_{n+1} = \epsilon_n$, since $3n!$ cannot divide 2. Thus one sign ϵ works at every level. Every entry of $A - \epsilon I$ is divisible by the unbounded sequence $3n!$, hence $A = \epsilon I$ and $[A] = 1$. $\square$

Theorem 3.2 now gives $\text{Per}(M_\infty) = \emptyset$ for this explicit cusped tower and quotient-order divergence for every hyperbolic owner.

The finite diagnostic freezes three matrices in $\Gamma(3)$ and eight moduli

$$3, 6, 18, 72, 360, 2160, 15120, 120960.$$

Their projective order sequences are

$$\begin{aligned} \text{G3-A} : & \quad 1, 3, 3, 6, 6, 36, 72, 288, \\ \text{G3-B} : & \quad 1, 1, 3, 12, 60, 360, 360, 2880, \\ \text{G3-C} : & \quad 1, 2, 6, 12, 12, 72, 72, 576. \end{aligned}$$

Sequential matrix multiplication and an independent finite-group-bound factor reduction agree in all 24 rows, and all 21 noninitial bonding transitions pass. These exact-integer rows illustrate the theorem,

including plateaus, but do not prove asymptotic divergence by extrapolation. Positive-word primitivity is checked; full $\Gamma(3)$ conjugacy primitivity for these three loop rows remains unestablished and is not needed for quotient orders.

The plateaus are informative. Divisibility permits $o_{n+1}(g) = o_n(g)$ for several successive levels, so monotone growth should not be read as strict growth. What the proof guarantees is eventual escape from every fixed bound. The finite table tests the projective-sign conventions, quotient arithmetic, and bonding maps on the registered rows; the asymptotic conclusion comes instead from Lemma 4.1 and Theorem 3.2. This separation prevents a short factorial schedule from being presented as empirical evidence for an infinite limit.

The two order algorithms provide a useful internal adversary. Direct multiplication can certify the first return but may be expensive, while factor reduction begins with a finite group-order bound and removes prime factors whenever the corresponding power is already projectively scalar. Their agreement checks both minimality and the sign quotient. It does not certify geodesic primitivity, which is a different owner question and is intentionally reported at a weaker level for these cusped diagnostics.

5 A cocompact residual control

Let Σ be a marked closed hyperbolic surface of genus two and $\Gamma = \pi_1(\Sigma)$. For $n \geq 1$, define R_n as the intersection of all normal subgroups of Γ of index at most n , let

$$H_n = \ker(\Gamma \rightarrow H_1(\Sigma; \mathbb{Z}/n!\mathbb{Z})), \quad \Gamma_n = R_n \cap H_n.$$

Proposition 5.1 (closed residual/homology tower). *The groups Γ_n form a descending normal finite-index residual tower. If the integral homology vector of g has content d , then*

$$\frac{n!}{\gcd(n!, d)} \mid o_n(g).$$

For content-one g , $n! \mid o_n(g)$. If g is represented by a closed geodesic and has primitive homology, that geodesic is primitive and its minimal lifted period is

$$T_n(g) = o_n(g)\ell(g) \geq n!\ell(g).$$

Proof. A finitely generated group has only finitely many subgroups of any fixed index, so R_n has finite index. Normality is built into the definition. Residual finiteness gives $\bigcap_n R_n = \{e\}$, and intersecting with H_n preserves trivial intersection. The factorial moduli make both chains descending.

The image of g in integral homology is a vector v of content d . Its additive order modulo $n!$ is $n!/\gcd(n!, d)$. Since the quotient by Γ_n maps onto the homology quotient, the order of $g\Gamma_n$ is divisible by this number. If v is primitive and $g = h^q$ with $q > 1$, then $v = q[h]$ is not primitive, a contradiction. Thus g is not a proper power. The cyclic axis stabilizer then identifies $o_n(g)\ell(g)$ with the exact minimal period of a lifted component. $\square$

The executable retains three content-one owners at $n = 1, \dots, 8$ and certifies the lower-bound sequence

$$1, 2, 6, 24, 120, 720, 5040, 40320.$$

It does not enumerate R_n or claim the full quotient orders were computed. This control removes cusps, principal congruence, and arithmetic-lattice provenance. The causal mechanism behind period escape is normal residuality with one common clock, not arithmetic specificity.

6 Coefficientwise same-owner escape

Let g be a fixed primitive owner and introduce the formal owner variable

$$x_g = e^{-s\ell(g)}.$$

At level n , its reciprocal unweighted factor under the physical clock is

$$E_{n,g}(x_g) = (1 - x_g^{o_n(g)})^{-1}.$$

Theorem 6.1 (fixed-prefix escape). *For every fixed integer $D \geq 0$, there is $n_0(g, D)$ such that*

$$E_{n,g}(x_g) = 1 \pmod{x_g^{D+1}}$$

for every $n \geq n_0(g, D)$. The same conclusion holds for any fixed finite panel of primitive owners. Finite level-dependent multiplicities and scalar weights cannot create a coefficient below degree $o_n(g)$.

Proof. The support of the geometric series is

$$0, o_n(g), 2o_n(g), \dots$$

Theorem 3.2 gives $o_n(g) \rightarrow \infty$, so eventually $o_n(g) > D$. A finite exponent or scalar changes only the coefficients on the same support. For a finite panel, take the maximum of the finitely many thresholds. Equivalently, the minimum lifted physical period in the panel tends to infinity. $\square$

This theorem strengthens the owner firewall. Not only are finite-level geodesics different from inverse-limit periodic points, but each fixed owner's formal Euler contribution becomes trivial on every fixed prefix if the original clock and owner identity are retained. It does not rule out a new collective candidate with a declared rescaled clock and normalization. Such a candidate must be registered separately, which is done next.

6.1 What fixed-prefix convergence does and does not say

The natural setting of Theorem 6.1 is the formal power-series ring in an owner variable. Fixed-prefix convergence means that for each D , the coefficients of degrees at most D eventually equal those of the constant series 1. It is stronger than observing that a factor is numerically close to one at a few chosen values of s , because every bounded formal degree is controlled at once. It is weaker than a global analytic convergence theorem: the theorem supplies no common domain, no bound uniform over an infinite owner family, and no justification for exchanging an inverse-limit operation with an infinite Euler product.

The variable x_g deliberately preserves owner identity. If several owners were prematurely projected to one scalar variable, equal or commensurable lengths could merge coefficients and obscure which factor produced them. The owner-indexed statement avoids that cancellation. After the same-owner result is proved, a finite panel may be specialized to numerical lengths because a maximum of finitely many escape thresholds still exists. A panel growing with n is different: newly introduced owners need not satisfy a uniform threshold, and the theorem makes no assertion about them.

Multiplicities also require precise wording. A finite exponent m_n at level n changes coefficients beginning at degree $o_n(g)$ but cannot create a lower-degree term. Hence arbitrary finite level-dependent lift counts leave fixed-prefix escape intact. Analytic norms can behave differently if m_n grows rapidly, since a large exponent can offset the smallness of $x_g^{o_n(g)}$ away from the formal origin. The four-quadrant analysis below is therefore conducted coefficientwise and exactly. It diagnoses support and multiplicity as separate axes rather than inferring one from the other.

7 Pure homology-cover calibrator

Retain the same marked genus-two surface, but now define the distinct tower

$$H_N = \ker(\Gamma \rightarrow H_1(\Sigma; \mathbb{Z}/N\mathbb{Z})).$$

Its deck group is $(\mathbb{Z}/N\mathbb{Z})^4$. This tower is not residual: its intersection is the commutator subgroup $[\Gamma, \Gamma]$. The owner panel consists of three primitive content-one classes, and the executed schedule is $N = n!$ for $n = 1, \dots, 8$.

Theorem 7.1 (lift order, count, and period). *For a primitive content-one owner g and every $N \geq 1$:*

- (i) *the cover degree is N^4 ;*
- (ii) *the deck image of g has order N ;*
- (iii) *the full preimage of the base geodesic has exactly N^3 primitive components; and*
- (iv) *every component has physical period $N\ell(g)$.*

Proof. The abelianization of the marked genus-two group is $\mathbb{Z}^4$, and reduction modulo N is onto $(\mathbb{Z}/N\mathbb{Z})^4$, proving the degree. Let $v \in \mathbb{Z}^4$ be the homology vector of g . Since its content is one, Bézout's identity gives an integer linear combination of its coordinates equal to one. Thus $kv = 0$ modulo N forces $N \mid k$, and the order is exactly N .

The lift of the base loop through a deck point follows translation by v . Its orbits have N deck points, so the N^4 deck points split into N^3 cycles. Moreover,

$$H_N \cap \langle g \rangle = \langle g^N \rangle.$$

Therefore a cycle has length $N\ell(g)$. To prove lift primitivity, suppose $g^N = h^q$ with $h \in H_N$ and $q \geq 2$. Centralizers of nontrivial elements in a torsion-free closed surface group are cyclic, so the unique-root property puts $h = g^r$. Then $rq = N$, whereas $h \in H_N$ forces $N \mid r$, which is impossible for $q \geq 2$. $\square$

The theorem also explains why this construction cannot be described as a renormalization of the residual inverse-limit owner without qualification. The tower H_N forgets all information in the commutator subgroup and therefore has nontrivial intersection. Its lifted components are indexed by an abelian deck group, and its clock will be explicitly divided by N . Those are changes to the tower and the time observable, not harmless reformulations. We retain the same base surface only to make the comparison controlled.

Content one is the exact hypothesis behind the clean powers of N . If the homology vector had content $d > 1$, its order modulo N would be $N / \gcd(N, d)$; both component count and lifted period would change with the arithmetic of N . Freezing primitive content-one owners removes that extra variable and makes the quadrant comparison isolate only clock and multiplicity. Primitive homology also certifies that the base geodesic is not a proper power, so the term “primitive component” is earned rather than inferred from a short word.

8 The four renormalization quadrants

Write $x_g = e^{-s\ell(g)}$. Compare the physical clock $N\ell(g)$ with the declared rescaled clock $\ell(g)$, and compare raw multiplicity N^3 with the declared logarithmic geometric mean

$$\log Z^{\text{ren}} = \frac{1}{N^3} \log Z_{\text{panel}}.$$

Theorem 8.1 (four-quadrant law). *For each owner and level, the four factors are exactly those in Table 1. Only the simultaneous clock and multiplicity renormalization recovers the base factor at every level.*

Table 1: Clock and multiplicity interventions for one owner.

Quadrant	clock	multiplicity	factor and behavior
Q_{00}	physical	raw	$(1 - x_g^N)^{-N^3} \rightarrow 1$ coefficientwise
Q_{10}	rescaled	raw	$(1 - x_g)^{-N^3}; [x_g] = N^3$
Q_{01}	physical	geometric mean	$(1 - x_g^N)^{-1} \rightarrow 1$ coefficientwise
Q_{11}	rescaled	geometric mean	$(1 - x_g)^{-1}$ exactly

Proof. Theorem 7.1 supplies N^3 identical lift factors. Physical time gives monomial x_g^N , while the $1/N$ time rescaling gives x_g . Raw multiplicity raises the factor to N^3 , and logarithmic normalization divides that exponent by N^3 . These choices yield the four displayed factors.

For Q_{00} and Q_{01} , the first nonconstant degree is N , so every fixed prefix is eventually one. For Q_{10} , the coefficient of x_g is N^3 and diverges. The Q_{11} expression is identically the base factor at every level. $\square$

For completeness, the coefficient calculation is exact:

$$[x^d](1 - x^m)^{-k} = \begin{cases} 0, & m \nmid d, \\ \binom{k+d/m-1}{d/m}, & m \mid d. \end{cases}$$

The finite replay contains 96 owner/level/quadrant rows and 1,248 integer coefficient rows through degree 12. It validates serialization and all displayed cases, but the theorem follows from the cover calculation for every N .

For a frozen panel P , the argument multiplies ownerwise. In independent variables one obtains

$$Z_{Q_{11}, N}(\{x_g\}_{g \in P}) = \prod_{g \in P} \left[(1 - x_g)^{-N^3} \right]^{1/N^3} = \prod_{g \in P} (1 - x_g)^{-1}.$$

Equivalently, the normalized logarithm is the arithmetic mean of the N^3 identical lift-component logarithms for each owner. No branch of a complex fractional power is needed: the definition is made at the logarithmic formal-series level, where division by N^3 is unambiguous and the constant term is zero.

This panel identity is exact for every N , including $N = 1$, but exactness alone does not make it canonical. The $1/N$ clock rescaling is chosen because the deck order of every registered owner is N , and the $1/N^3$ logarithmic normalization is chosen because the number of components is N^3 . Both are read from the desired base owner through the marked homology cover. An arithmetic construction would still have to explain why these interventions arise intrinsically and why an infinite primitive product converges. The present theorem instead records the full cost of making a finite target factor reappear.

Proposition 8.2 (uniqueness within scalar clock/exponent renormalizations). *Fix N and one owner. Let $c_N > 0$ multiply the physical lifted period and let $b_N > 0$ multiply the logarithm of the raw N^3 -component product. If*

$$\left(1 - e^{-sc_N N \ell(g)}\right)^{-b_N N^3} = \left(1 - e^{-s \ell(g)}\right)^{-1}$$

for every sufficiently large real s , then

$$c_N = \frac{1}{N}, \quad b_N = \frac{1}{N^3}.$$

Proof. Put $q = e^{-s \ell(g)}$. Taking logarithms gives

$$b_N N^3 \sum_{r \geq 1} \frac{q^{rc_N N}}{r} = \sum_{r \geq 1} \frac{q^r}{r}.$$

As $q \downarrow 0$, the smallest positive exponent on the left is $c_N N$, while that on the right is one. Equality on an interval forces $c_N N = 1$. The coefficient of the resulting leading term then forces $b_N N^3 = 1$. $\square$

Thus Q_{11} is not merely one successful corner among arbitrary scalar choices: within this registered two-parameter intervention class, its clock and exponent are forced by all- s equality. The proposition does not exclude owner-dependent weights, mixtures between owners, nonlocal transforms, or operator-valued renormalizations. Any such construction would be a further candidate with additional degrees of freedom and would need new controls. The current result keeps the intervention class small enough that the positive equality remains auditable.

The factorial schedule $N = n!$ used by the executable is likewise a finite validation schedule rather than part of Proposition 8.2. The cover theorem and uniqueness argument hold for every positive integer N . Factorials are retained only to compare the calibrator on the same eight-level scale as the closed residual/homology control and to make divisibility between displayed levels explicit.

9 Computational certificates

The residual-tower, closed-control, and homology-calibrator artifacts use deterministic Python-standard-library code. The Round-8 calibrator is frozen by SHA-256

88d10c3dcdee3387b16414d2c56d4934b6daeeef6728acc689855049840850a72.

Its locked Round-5 owner ledger and validation hashes are

0c74333b63f6027b16d134f19a320b8148e7fab6f86fa204d213c801106fe825,
afdc51ca7ecfbd8777955c7438f08d4580e6b924419a807191e097b0292d9c10.

Twelve unit tests check the cover formulas, the content-one condition, every quadrant, exact coefficients, owner separation, and Route boundaries. Two isolated builds are byte-identical. The Round-8 core hash is

a1b588724dacb2ab2986326a7a5e1c6aec654c61538c1465e26564357b568b33.

The default command `bash experiments/reproduce_round8.sh` verifies the checked-in files without modifying them; `--refresh` is explicit.

The finite replay is an exact executable audit. Its evidence status is

NUMERICALLY_CERTIFIED.

The cover and factor theorems have status **PROVED** from the arguments above. These labels are not interchangeable.

The executable certificate mirrors the proof chain. It first validates the immutable owner registry and the content-one homology vectors. It then derives cover degree, deck order, component count, physical and rescaled periods, and the four symbolic factor descriptors. Coefficients are computed as integers from the binomial formula rather than estimated by evaluating a product at floating-point values. Row counts and uniqueness constraints make omissions and duplicates fail closed. Finally, two isolated output trees are compared bitwise and bound to the checked-in hashes.

The earlier residual and closed-control artifacts serve different roles and are not mixed into the positive calibrator. The congruence rows check projective quotient orders and bonding divisibility; the closed rows check the factorial homology lower bound; the Round-8 rows check the four homology-cover quadrants. This separation preserves candidate identity across computational stages. Reproducing all ledgers supports the finite claims and their serialization, while the universal statements in Theorems 3.2, 6.1, 7.1, and 8.1 rest on the written proofs.

10 Adversarial and Route-A analysis

The original residual inverse-limit candidate has weak arithmetic provenance in its congruence specialization but no periodic points. Its formal tuple is

(A0_WEAK_ARITHMETIC_RELATION, A1_FAIL, A2_FAIL, A3_FAIL, A4_FAIL),

with `ROUTE_A_REJECTED`. A1 fails analytically because $\text{Per}(M_\infty) = \emptyset$. A2 is not a failed numerical divisor match; it is not testable on primitive periodic orbits of this owner because there are none.

The pure homology-cover panel is a new candidate with new tower, clock, normalization, and owner declaration. Its exact finite-panel orbit ledger supports `A1_PASS_ANALYTIC`, but A0 fails: the construction is generic for every marked genus-two hyperbolic metric and contains no intrinsic rational-prime correspondence or natural $\log p$ scale. The same proof works for nonarithmetic controls. This is the decisive proves-too-much test. A2 fails because no full primitive census or global determinant is defined; A3 and A4 fail because there is no global analytic structure or natural operator lift. Its tuple is

$$(A0_FAIL, A1_PASS_ANALYTIC, A2_FAIL, A3_FAIL, A4_FAIL),$$

again `ROUTE_A_REJECTED`. Route B remains unauthorized for both objects.

The four quadrants also function as internal adversarial controls. Q_{10} shows that support repair does not control lift multiplicity; Q_{01} shows that multiplicity repair does not control support. The positive Q_{11} equality is therefore mathematically precise but arithmetically nonselective. It cannot inherit Route credit from the rejected residual owner.

11 Limitations and open obligations

The no-period theorem assumes a normal residual tower and one common clock. It does not cover arbitrary nonnormal chains without additional conjugator control. The three cusped diagnostic rows lack full conjugacy-primitivity certificates, so their computed times are loop-closing diagnostics rather than asserted primitive minimal periods. The closed control avoids that issue through primitive homology, but its full residual-core quotient orders are not enumerated.

The renormalized identity concerns a fixed finite owner panel. The pure homology tower is not residual, and no interchange between a tower limit and an infinite product over all primitive geodesics is justified. Growing panels could introduce uncontrolled orbit proliferation and would require a new convergence theorem. The logarithmic $1/N^3$ normalization is explicitly imposed from deck multiplicity; it is not derived from arithmetic weights. Finally, no target divisor, functional equation, transfer operator, or quantum lift appears. These limitations block any promotion from a finite calibration identity to a Hilbert–Pólya claim.

There is no contradiction between the negative residual-tower theorem and the positive homology-cover identity. The former fixes a coherent inverse-limit point, a residual tower, and the physical clock; the latter fixes a base-owner panel, replaces the tower by a nonresidual abelian cover, rescales time at each level, and averages the logarithms of distinct lift components. Since both the quantified objects and the observables change, Q_{11} does not restore periodic points to M_∞ . It constructs a separate finite factor identity.

A natural next question is whether a canonically weighted growing panel admits a locally uniform logarithmic limit after both interventions. Answering it would require at least a primitive-owner counting law, control of lift multiplicities uniformly over that population, and bounds that justify summing before taking the tower limit. None of those inputs is contained in the 96-row calibrator. Coefficientwise equality for each fixed finite panel does not supply uniformity as the panel grows.

12 Conclusion

Residual towers erase same-owner periodicity in two compatible senses: the inverse-limit flow has no periodic point, and every fixed finite Euler panel escapes every fixed physical-time or coefficient window. A new homology-cover candidate can undo this effect, but only by changing both clock and logarithmic multiplicity. The four-quadrant theorem isolates those two costs exactly. Because the recovered identity is finite-panel and generic, its proper role is calibration: it shows how easy it is to manufacture an exact tower identity that contains no arithmetic specificity. This owner discipline is the paper’s main methodological result and the reason neither construction advances to Route-A A2 or Route B.

Data and Code Availability

All data are repository artifacts under `papers/27-congruence-inverse-limit-no-go/results`. Deterministic source code, tests, freeze contracts, validation reports, and verify-default reproduction scripts are available in the corresponding `code` and `experiments` directories. No proprietary, personal, prime-target, or Riemann-zero dataset was used.

Ethics Declaration

This theoretical and computational mathematics study involved no human participants, personal data, animals, clinical material, or field intervention. Institutional ethics approval and informed consent were not applicable.

Author Contributions

Liang Wang: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing—Original Draft, Writing—Review & Editing, Visualization, and Project Administration.

Conflict of Interest

The author declares no conflict of interest.

Funding

This research received no external funding.

AI-Assisted Research Disclosure

AI-assisted tools were used to support research organization, source-metadata checking, code development, mathematical exposition, and manuscript drafting. The author retains responsibility for the definitions, theorem statements, proofs, computations, citations, claim boundaries, and final text. Exact computational claims are accompanied by deterministic tests, source locks, and reproducibility receipts; AI output is not treated as independent evidence.

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